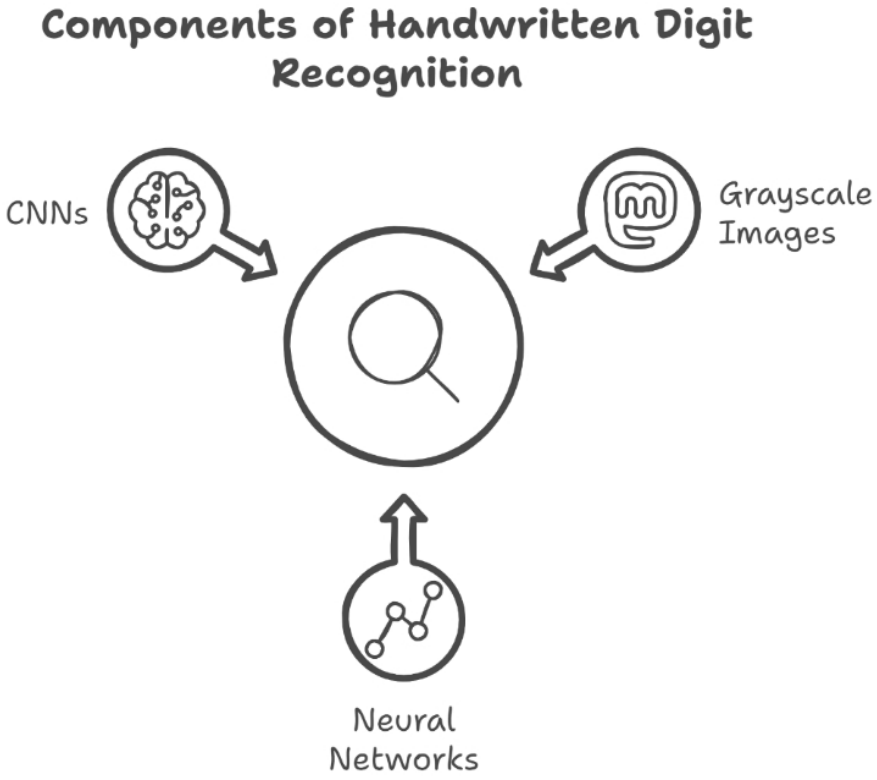


Characteristics of the MNIST Dataset

- **Image Size**: Each image is 28x28 pixels.
- **Grayscale Values**: Pixel values range from 0 to 255, where 0 represents black and 255 represents white.
- **Classes**: There are 10 classes (digits 0 through 9).
- **Training Set**: 60,000 images.
- **Test Set**: 10,000 images.

Objective

The goal is to classify each image into one of the 10 digit classes using a convolutional neural network (CNN).

Implementation

Below is a step-by-step implementation using TensorFlow and Keras.

Step 1: Import Libraries